

ASSIGNMENT NO.2: Basic Data Retrieval using SQL

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Q1) Create a database named CollegeDB and a table named Student.

The table should contain the following columns:

Column Name	Data Type
StudentID	INT
StudentName	VARCHAR(50)
Age	INT
Course	VARCHAR(30)
City	VARCHAR(30)
Marks	INT

Add a primary key on StudentID and insert at least 10 records.

QUERY:

```
1 CREATE DATABASE CollegeDB;
2 USE CollegeDB;
3
4 CREATE TABLE Student (
5     StudentID INT PRIMARY KEY,
6     StudentName VARCHAR(50),
7     Age INT,
8     Course VARCHAR(30),
9     City VARCHAR(30),
10    Marks INT
11 );
12 INSERT INTO Student
13 VALUES
14 (1,'Aarav Shah',18,'BSc IT','Mumbai',85),
15 (2,'Meera Joshi',19,'BCA','Pune',78),
16 (3,'Rohan Patil',20,'BSc CS','Nashik',88),
17 (4,'Sneha Desai',18,'BCA','Mumbai',92),
18 (5,'Amit Kumar',21,'BSc IT','Delhi',75),
19 (6,'Pooja Singh',19,'BSc CS','Pune',81),
20 (7,'Vikram Rao',20,'BCA','Bengaluru',89),
21 (8,'Neha Sharma',18,'BSc IT','Nashik',94),
22 (9,'Rahul Verma',21,'BSc CS','Mumbai',73),
23 (10,'Kavita Iyer',19,'BCA','Chennai',87);
24
25 SELECT * FROM Student;
```

OUTPUT:

	StudentID	StudentName	Age	Course	City	Marks
☐ Edit ☐ Copy ☐ Delete	1	Aarav Shah	18	BSc IT	Mumbai	85
☐ Edit ☐ Copy ☐ Delete	2	Meera Joshi	19	BCA	Pune	78
☐ Edit ☐ Copy ☐ Delete	3	Rohan Patil	20	BSc CS	Nashik	88
☐ Edit ☐ Copy ☐ Delete	4	Sneha Desai	18	BCA	Mumbai	92
☐ Edit ☐ Copy ☐ Delete	5	Amit Kumar	21	BSc IT	Delhi	75
☐ Edit ☐ Copy ☐ Delete	6	Pooja Singh	19	BSc CS	Pune	81
☐ Edit ☐ Copy ☐ Delete	7	Vikram Rao	20	BCA	Bengaluru	89
☐ Edit ☐ Copy ☐ Delete	8	Neha Sharma	18	BSc IT	Nashik	94
☐ Edit ☐ Copy ☐ Delete	9	Rahul Verma	21	BSc CS	Mumbai	73
☐ Edit ☐ Copy ☐ Delete	10	Kavita Iyer	19	BCA	Chennai	87

Q2) Write a MySQL query to display all records from the Student table.

The output should display all columns and all rows.

QUERY:

```
1 SELECT * FROM Student;
```

OUTPUT:

	StudentID	StudentName	Age	Course	City	Marks
☐ Edit ☐ Copy ☐ Delete	1	Aarav Shah	18	BSc IT	Mumbai	85
☐ Edit ☐ Copy ☐ Delete	2	Meera Joshi	19	BCA	Pune	78
☐ Edit ☐ Copy ☐ Delete	3	Rohan Patil	20	BSc CS	Nashik	88
☐ Edit ☐ Copy ☐ Delete	4	Sneha Desai	18	BCA	Mumbai	92
☐ Edit ☐ Copy ☐ Delete	5	Amit Kumar	21	BSc IT	Delhi	75
☐ Edit ☐ Copy ☐ Delete	6	Pooja Singh	19	BSc CS	Pune	81
☐ Edit ☐ Copy ☐ Delete	7	Vikram Rao	20	BCA	Bengaluru	89
☐ Edit ☐ Copy ☐ Delete	8	Neha Sharma	18	BSc IT	Nashik	94
☐ Edit ☐ Copy ☐ Delete	9	Rahul Verma	21	BSc CS	Mumbai	73
☐ Edit ☐ Copy ☐ Delete	10	Kavita Iyer	19	BCA	Chennai	87

Q3) Write a MySQL query to display only the following columns:

1. StudentName
2. Course
3. City

QUERY:

```
1 SELECT StudentName, Course, City
2 FROM Student;
```

OUTPUT:

				StudentName	Course	City
☐		Edit		Copy		Delete
				Aarav Shah	BSc IT	Mumbai
☐		Edit		Copy		Delete
				Meera Joshi	BCA	Pune
☐		Edit		Copy		Delete
				Rohan Patil	BSc CS	Nashik
☐		Edit		Copy		Delete
				Sneha Desai	BCA	Mumbai
☐		Edit		Copy		Delete
				Amit Kumar	BSc IT	Delhi
☐		Edit		Copy		Delete
				Pooja Singh	BSc CS	Pune
☐		Edit		Copy		Delete
				Vikram Rao	BCA	Bengaluru
☐		Edit		Copy		Delete
				Neha Sharma	BSc IT	Nashik
☐		Edit		Copy		Delete
				Rahul Verma	BSc CS	Mumbai
☐		Edit		Copy		Delete
				Kavita Iyer	BCA	Chennai

Q4) Write a MySQL query to display students who belong to the city Mumbai. The output should include:

1. StudentID
2. StudentName
3. City
4. Marks

QUERY:

```
1 SELECT StudentID, StudentName, City, Marks
2 FROM Student
3 WHERE City = 'Mumbai';
```

OUTPUT:

				StudentID	StudentName	City	Marks
☐		Edit		Copy		Delete	
				1	Aarav Shah	Mumbai	85
☐		Edit		Copy		Delete	
				4	Sneha Desai	Mumbai	92
☐		Edit		Copy		Delete	
				9	Rahul Verma	Mumbai	73

Q5) Write a MySQL query to display students whose marks are greater than 80. The output should be sorted by marks in descending order.

QUERY:

```
1 SELECT *
2 FROM Student
3 WHERE Marks > 80
4 ORDER BY Marks DESC;
```

OUTPUT:

	StudentID	StudentName	Age	Course	City	Marks
☐ Edit Copy Delete	8	Neha Sharma	18	BSc IT	Nashik	94
☐ Edit Copy Delete	4	Sneha Desai	18	BCA	Mumbai	92
☐ Edit Copy Delete	7	Vikram Rao	20	BCA	Bengaluru	89
☐ Edit Copy Delete	3	Rohan Patil	20	BSc CS	Nashik	88
☐ Edit Copy Delete	10	Kavita Iyer	19	BCA	Chennai	87
☐ Edit Copy Delete	1	Aarav Shah	18	BSc IT	Mumbai	85
☐ Edit Copy Delete	6	Pooja Singh	19	BSc CS	Pune	81

Q6) Write a MySQL query to display the top 3 students based on marks.

The query should:

1. Display StudentName and Marks.
2. Sort marks from highest to lowest.
3. Display only 3 records using LIMIT.

QUERY:

```
1 SELECT StudentName, Marks
2 FROM Student
3 ORDER BY Marks DESC
4 LIMIT 3;
```

OUTPUT:

	StudentName	Marks
☐ Edit Copy Delete	Neha Sharma	94
☐ Edit Copy Delete	Sneha Desai	92
☐ Edit Copy Delete	Vikram Rao	89